COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling for Data Science

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Resampling Methods

Reading:
Chapter 5 from
"An Introduction to Statistical Learning"

Lesson Goals

- Understand the basic concepts of resampling methods for model selection and assessment.
- Understand the similarities and differences between the validation set approach, leave-one-out cross-validation, and K-fold cross-validation.
- Understand the basic theoretical concepts of the bootstrap method for assessing statistical accuracy.
- Recognize that the performance of machine learning models depends on their prediction capability on independent test data.

Resampling Methods

- Involves repeatedly drawing samples from a training set and refitting a model or estimator of interest on each sample to obtain more information about the fitted model/estimator.

- <u>Example</u>: We can estimate the variability of a linear regression fit by repeatedly drawing different samples from the training data, fitting a OLS regression to each new sample, and then examining the extent to which the resulting fits differ.

- *Model Assessment:* having chosen a final model, estimating its prediction error on new data.
- *Model Selection:* estimating the performance of different models in order to choose the best one.

Resampling Methods (cont.)

- *Cross-Validation*
  - Used to estimate test set prediction error rates associated with a given statistical model, or to select the appropriate level of model complexity.
- *Bootstrap*
  - Used mostly to provide a measure of accuracy of a parameter estimate or of a given statistical method.

Model Assessment

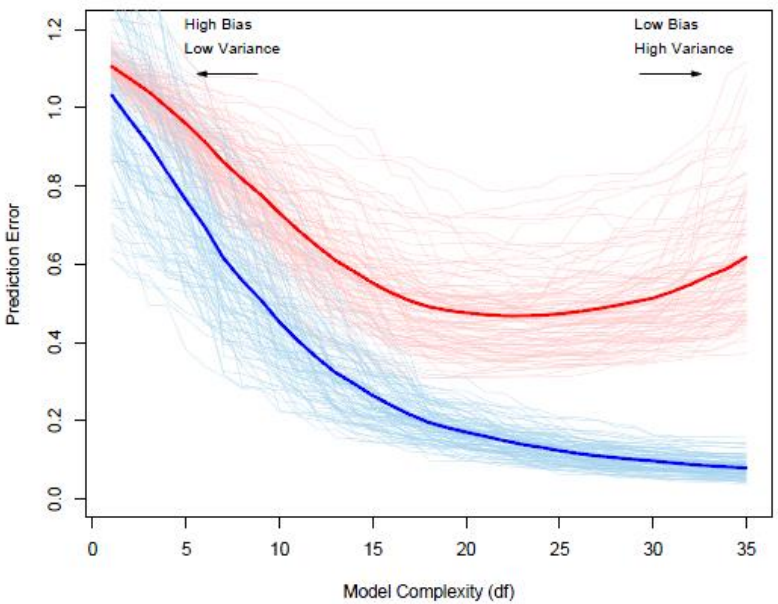
- *Test Error*
  - The average error that results from using a statistical method to predict the response on a <u>new</u> observation.
  - The prediction error over an independent test sample.

- *Training Error*
  - The average loss over the training sample:

$$\frac{1}{N}\sum_{i=1}^{N} L[y_i, \hat{f}(x_i)]$$

- **Note:** The training error rate can dramatically *underestimate* the test error rate

Model Assessment (cont.)



- As the model becomes more and more complex, it uses the training data more and can adapt to more complicated underlying structures.
- Hence, there is a decrease in bias but an increase in variance.
- However, training error is not a good estimate of the test error.

- Training error consistently decreases with model complexity.
- A model with zero training error is overfit to the training data and will typically generalize poorly.

Model Assessment (cont.)

- If we are in a data-rich situation, the best approach for both *model selection* and *model assessment* is to randomly divide the dataset into three parts: training set, validation set, and test set.

- The *training set* is used to fit the models. The *validation set* is used to estimate prediction error for model selection. The *test set* is used for assessment of the prediction error of the final chosen model.

- A typical split might by 50% for training, and 25% each for validation and testing.

Model Assessment (cont.)

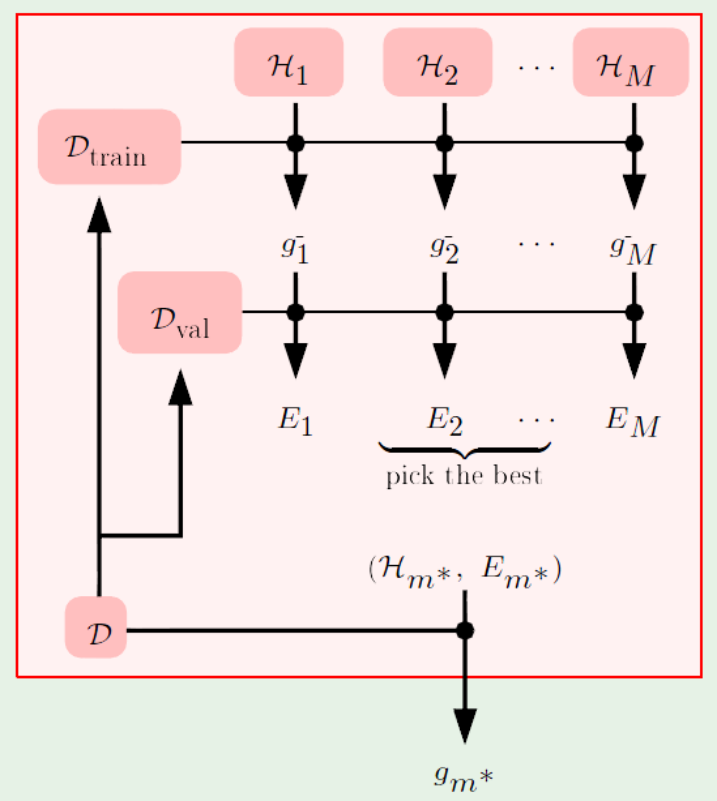
- Best solution: use a large designated test set, which is often not available. For the methods presented here, there is insufficient data to split it into three parts.

- There are some methods to make mathematical adjustments to the training error rate in order to estimate the test error rate:
  - C_p statistic, AIC, BIC (we will discuss these next week)

- Here, we consider cross-validation (CV) methods that estimate the test error by *holding out* a subset of the training observations from the fitting process, and then applying statistical method to those held out observations.
- For example, leave out 1/10 of subjects, develop model in the remaining 9/10, evaluate in the withheld 1/10, repeat 10 times and average (so called 10-fold CV)
- Still not very precise way to measure accuracy

Overview: Model Selection

- By far, the most important use of validation is for *model selection*.
- Suppose we have M models; validation can be used to select one of these models.
- We use the training data to fit the model, and we evaluate each model on the validation set to obtain the validation errors.
- It is now a simple matter to select the model with the lowest validation error.

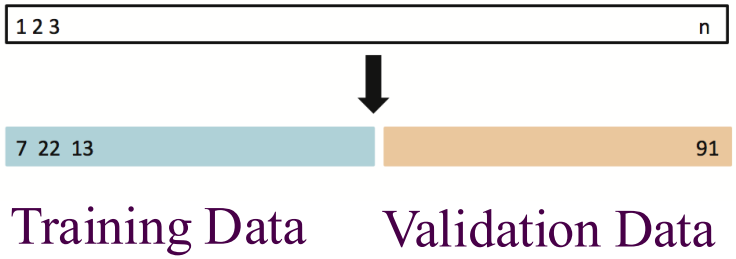


Validation Set Approach

- Suppose that we would like to find a set of variables that give the lowest *validation error rate* (an estimate of the *test error rate*).

- If we have a large data set, we can achieve this goal by randomly splitting the data into separate training and validation data sets.

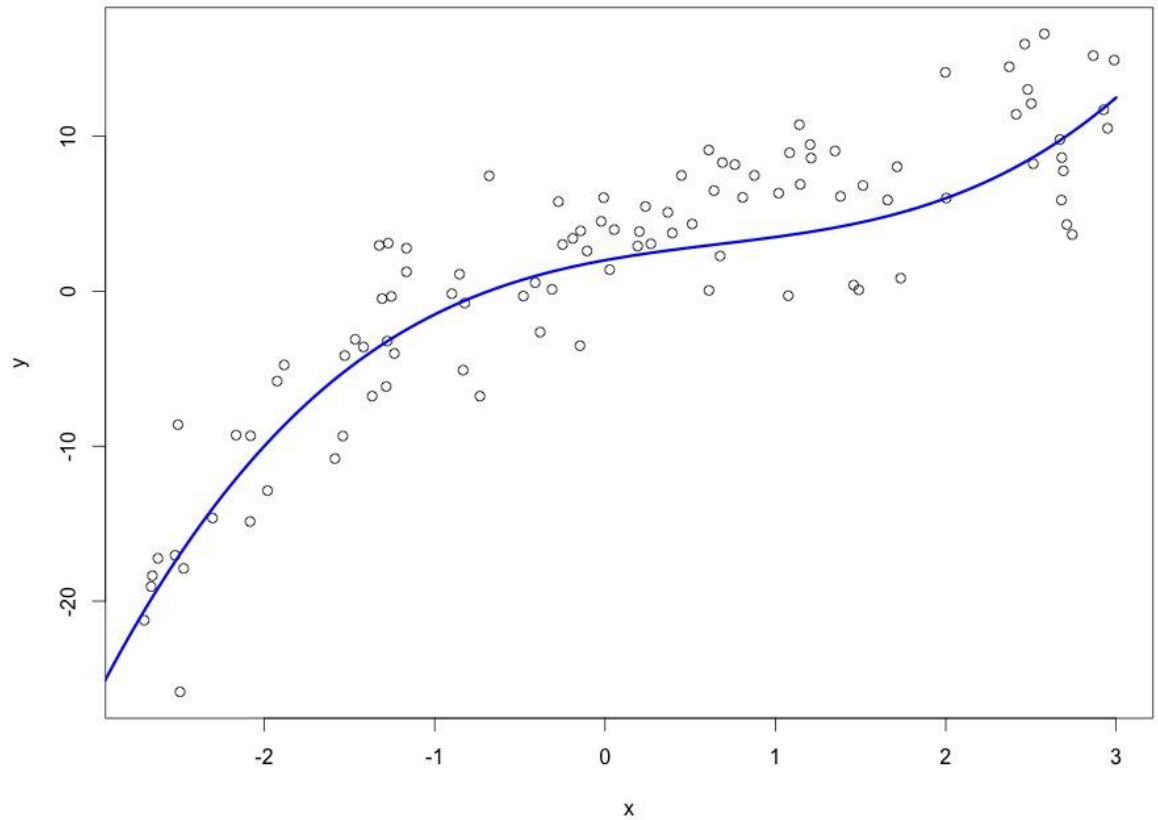
- Then, we use the training data set to build each possible model and select the model that gives the lowest error rate when applied to the validation data set.



Model Selection via Validation Set

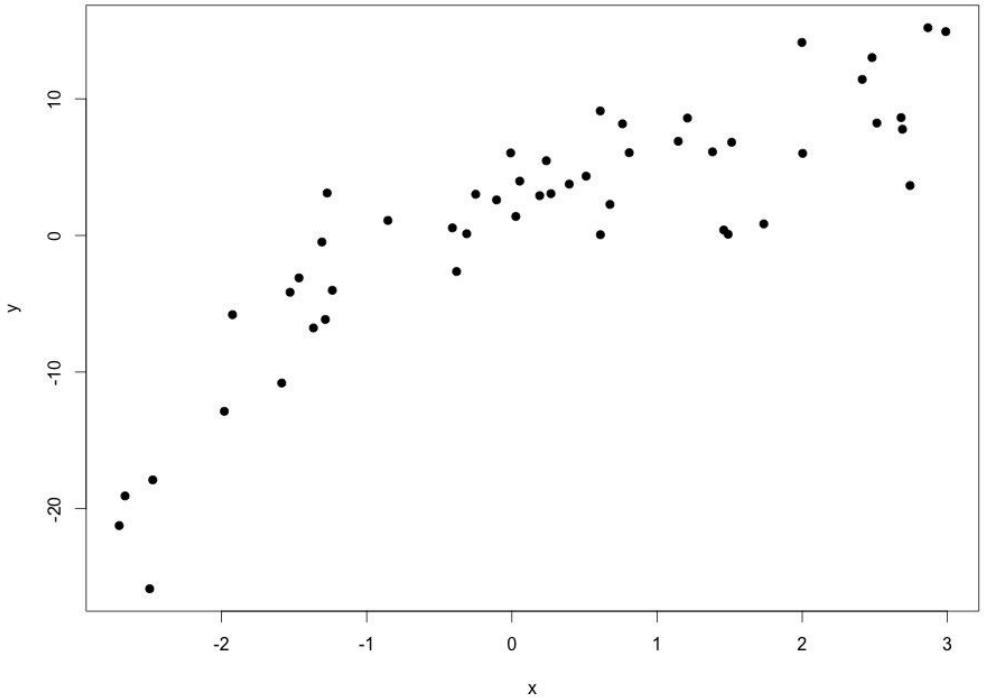
```
set.seed(3249)
x <- runif(100, -3, 3)
y <- 2 + 2*x - 1*x^2 + .5*x^3 + rnorm(100, mean = 1, sd = 4)
dat <- data.frame(cbind(x, y))
plot(dat)
curve(expr = 2 + 2*x - 1*x^2 + .5*x^3,
      from = -3, to = 3, n = 501, add = TRUE,
      lwd = 3, col = 4)
```

Generate data (100 cases) from a cubic model with normal errors. There is one outcome *y* and one predictor *x*.

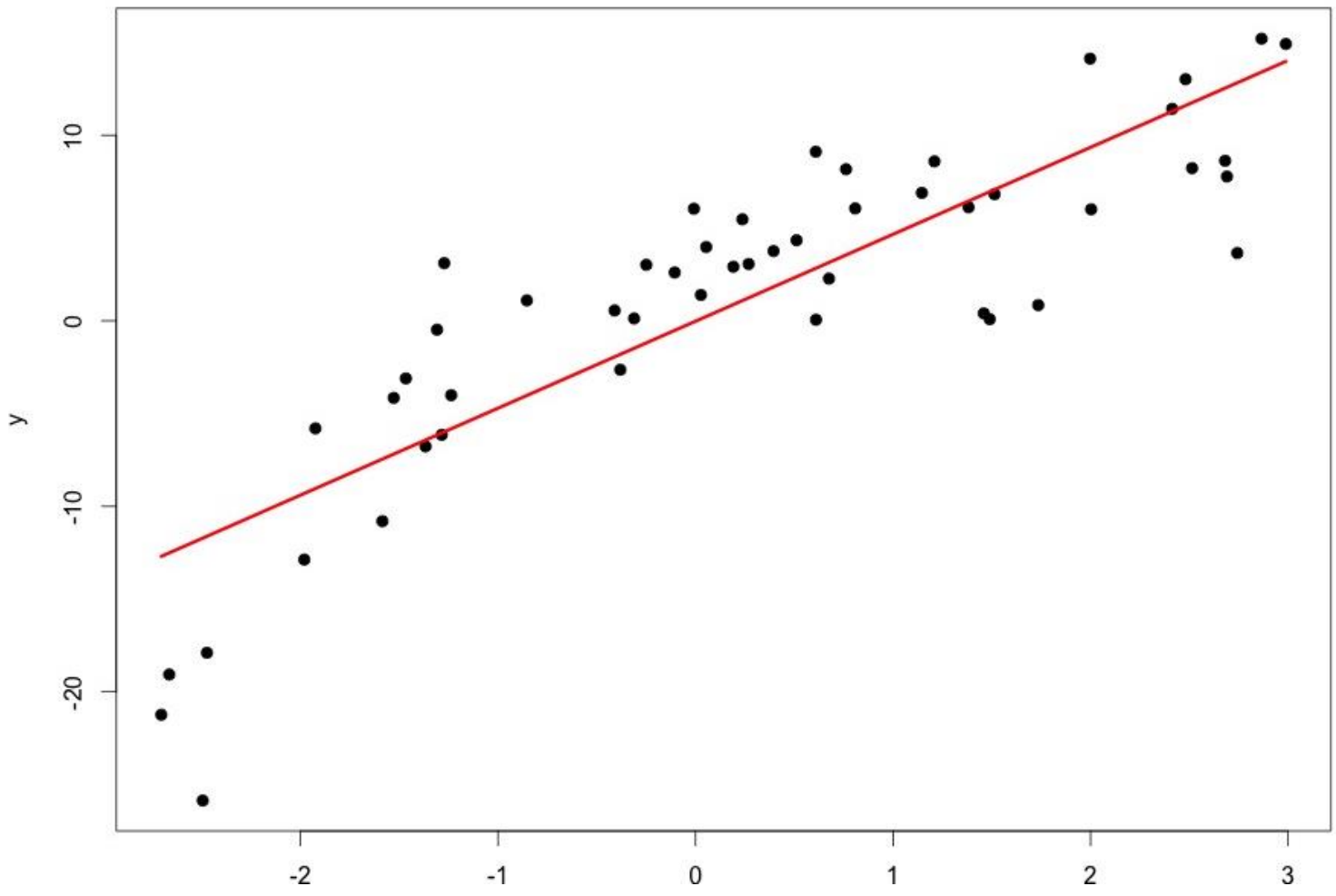


Model Selection via Validation Set

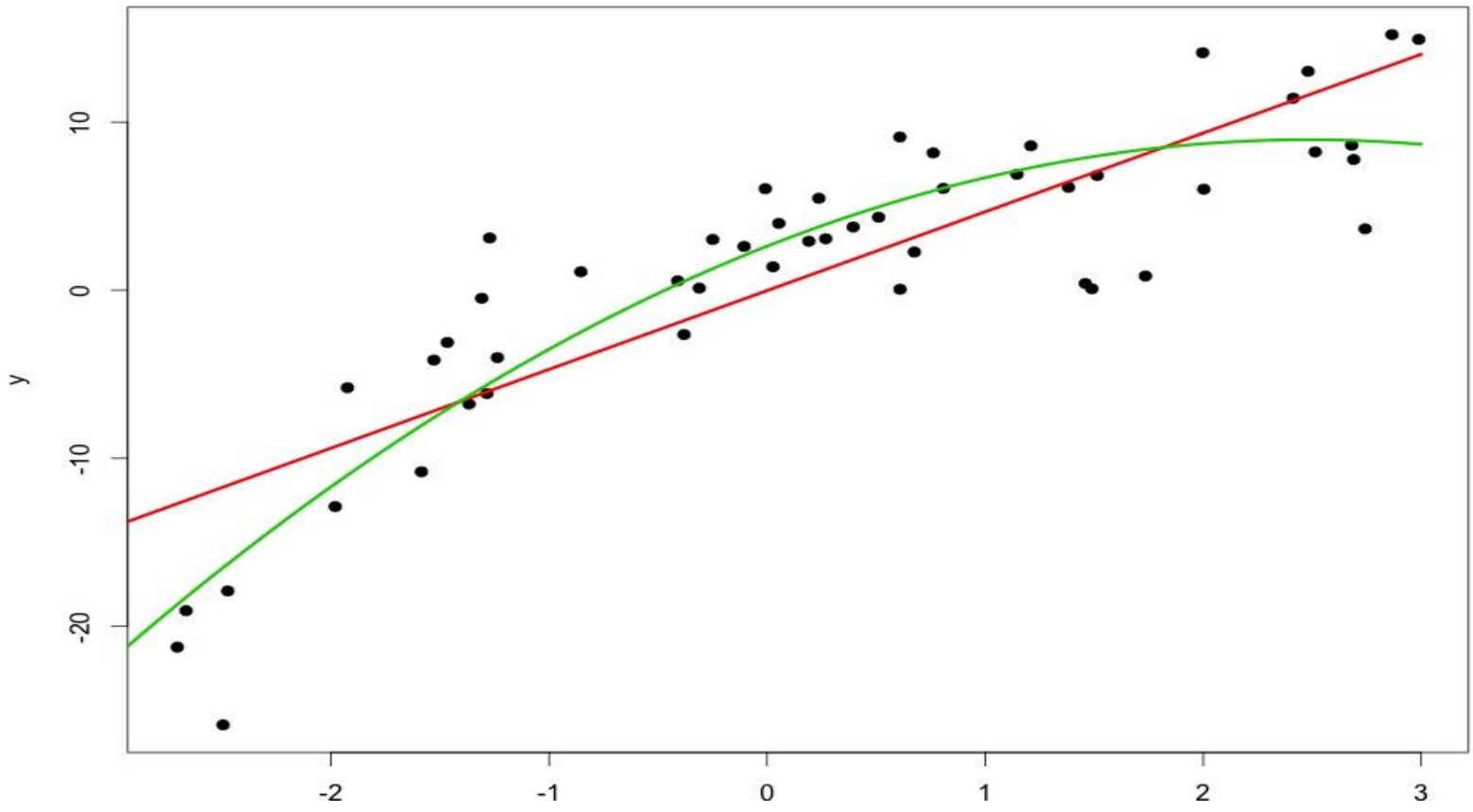
```
### Randomly select 1/2 of cases for a test set
set.seed(3487)
ind <- sample(rep(0:1, each = 50), replace = FALSE)
table(ind)
ind

train <- dat[ind == 0,]
str(train)
test <- dat[ind == 1,]
str(test)


### Plot the train set (half the data)
plot(train, pch = 19)
```



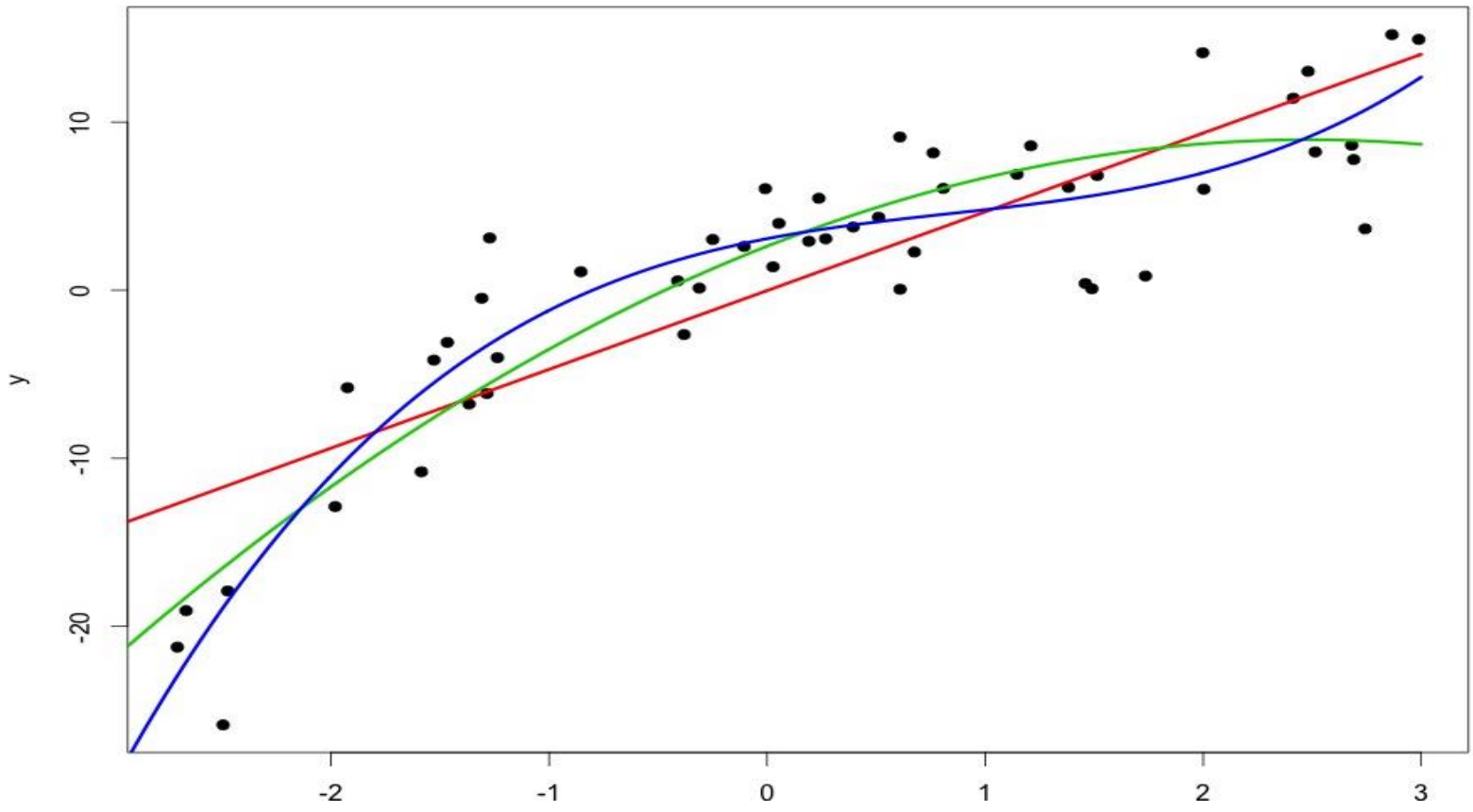
Polynomial Curve-Fitting Example



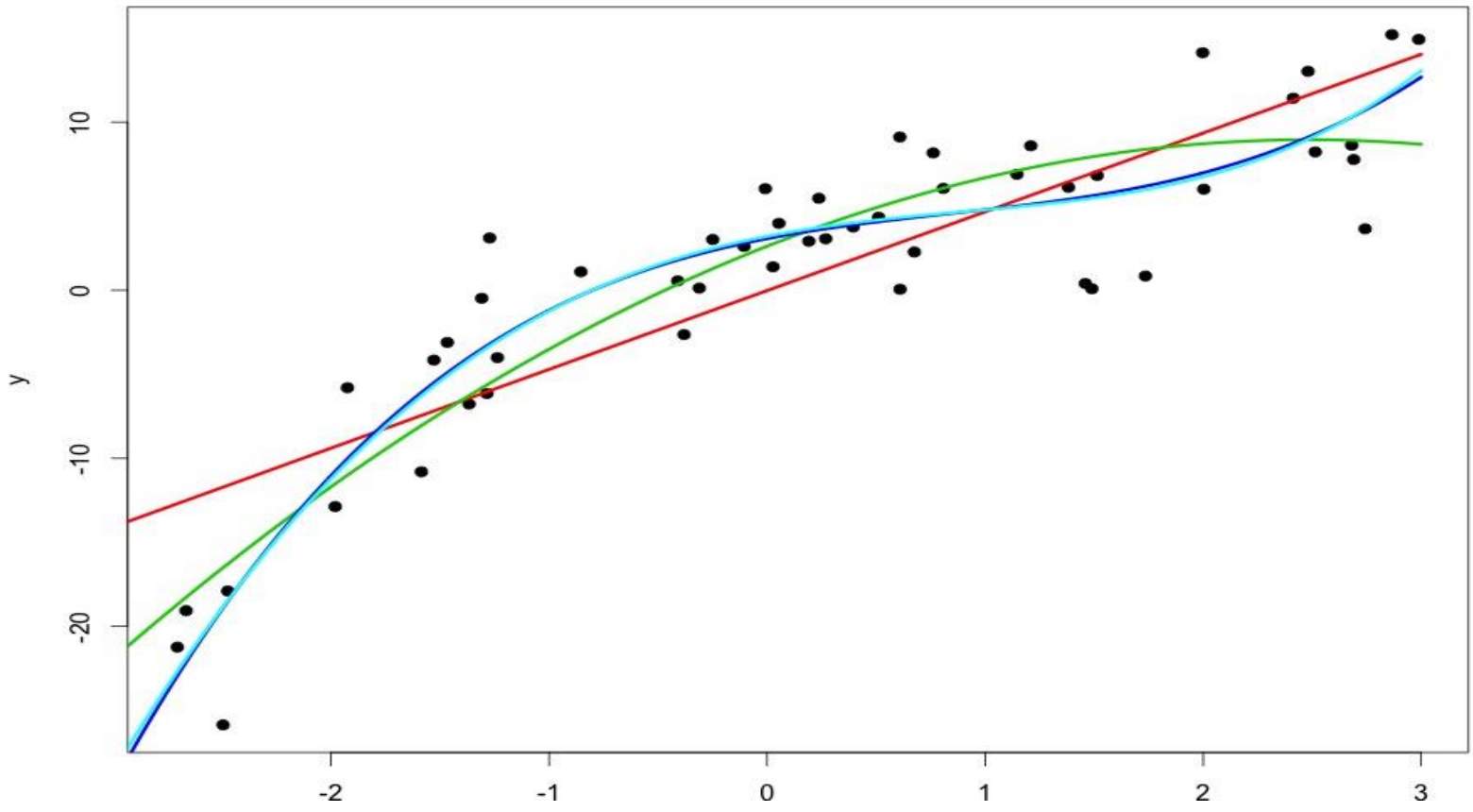
Polynomial Curve-Fitting Example



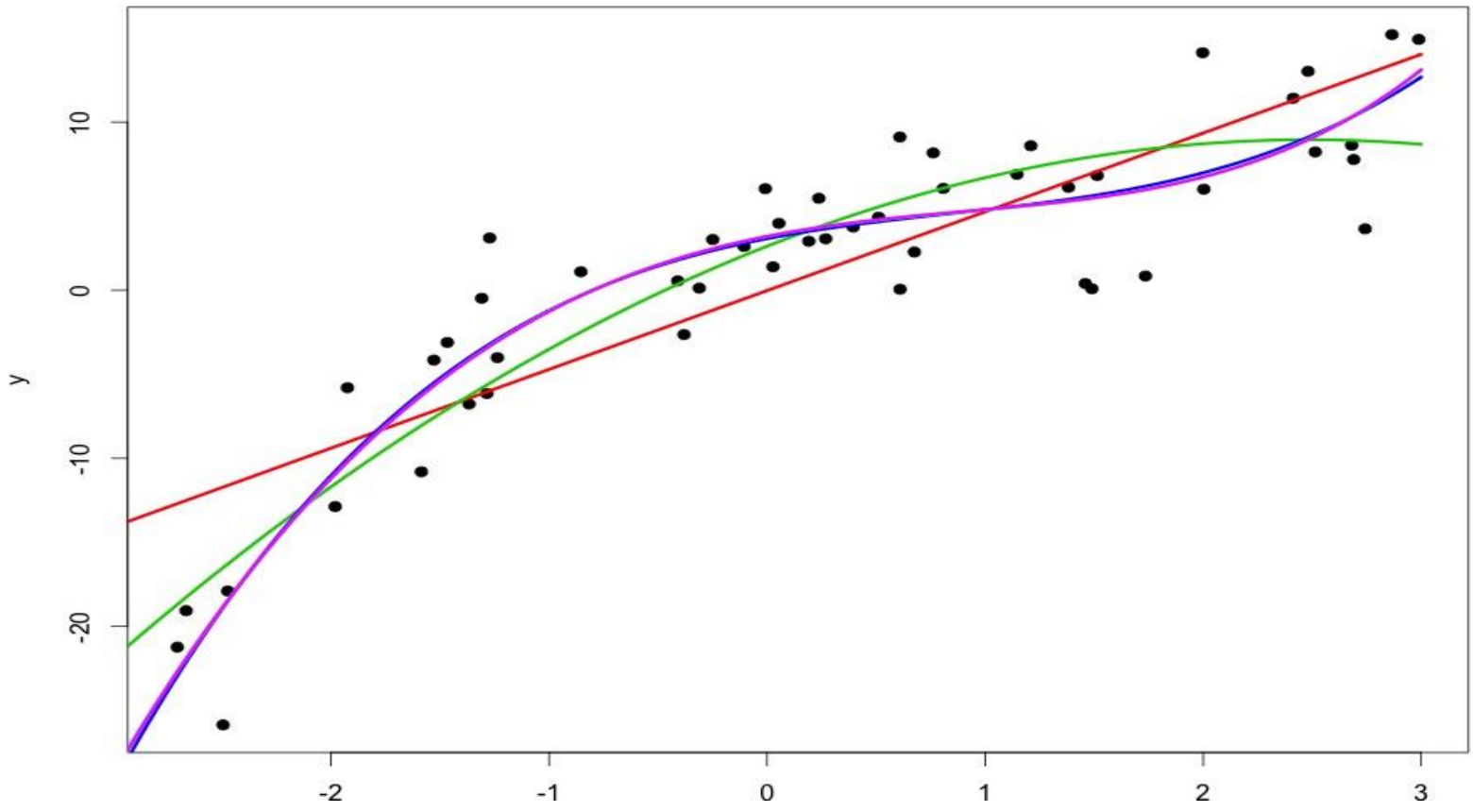
Polynomial Curve-Fitting Example



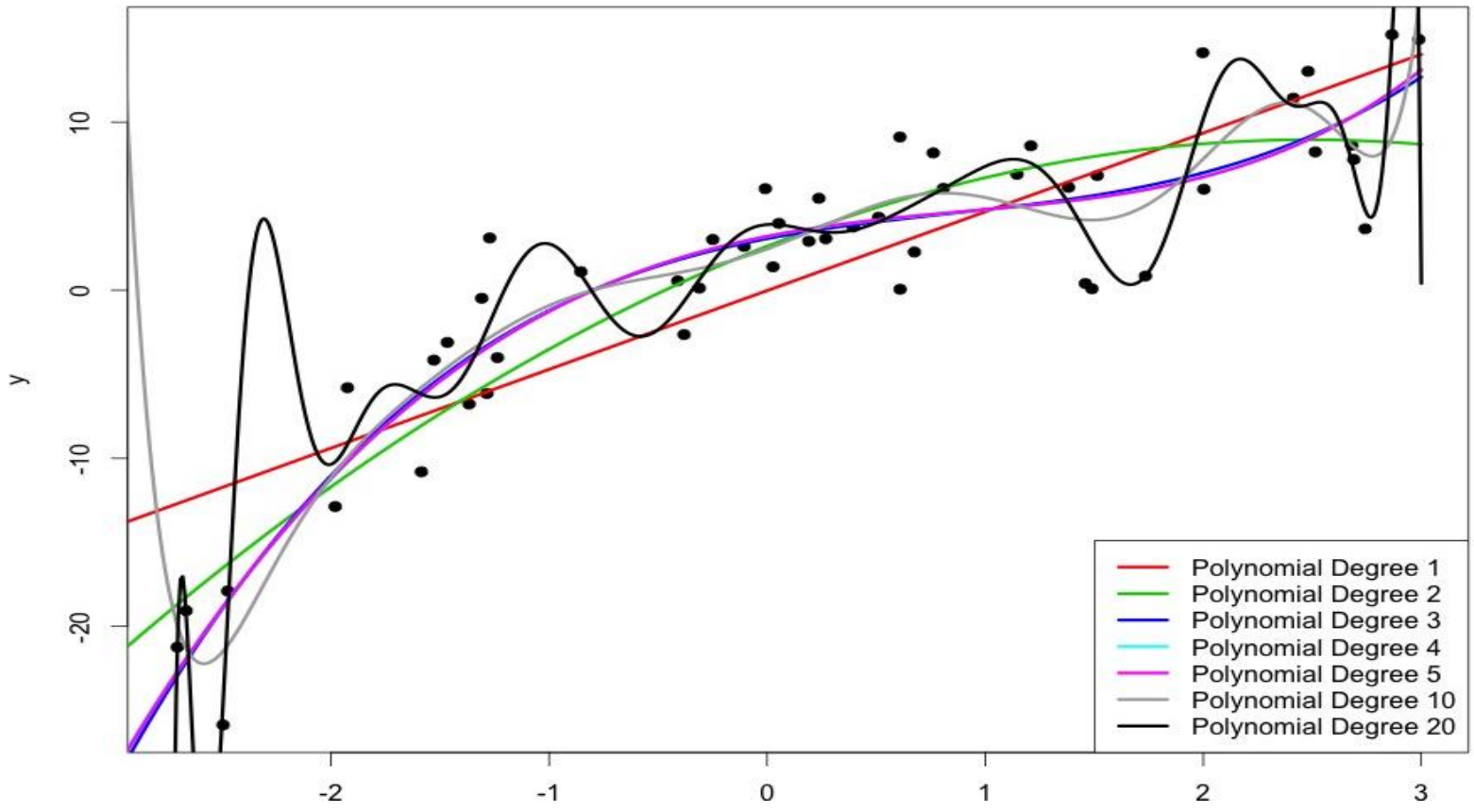
Polynomial Curve-Fitting Example



Polynomial Curve-Fitting Example

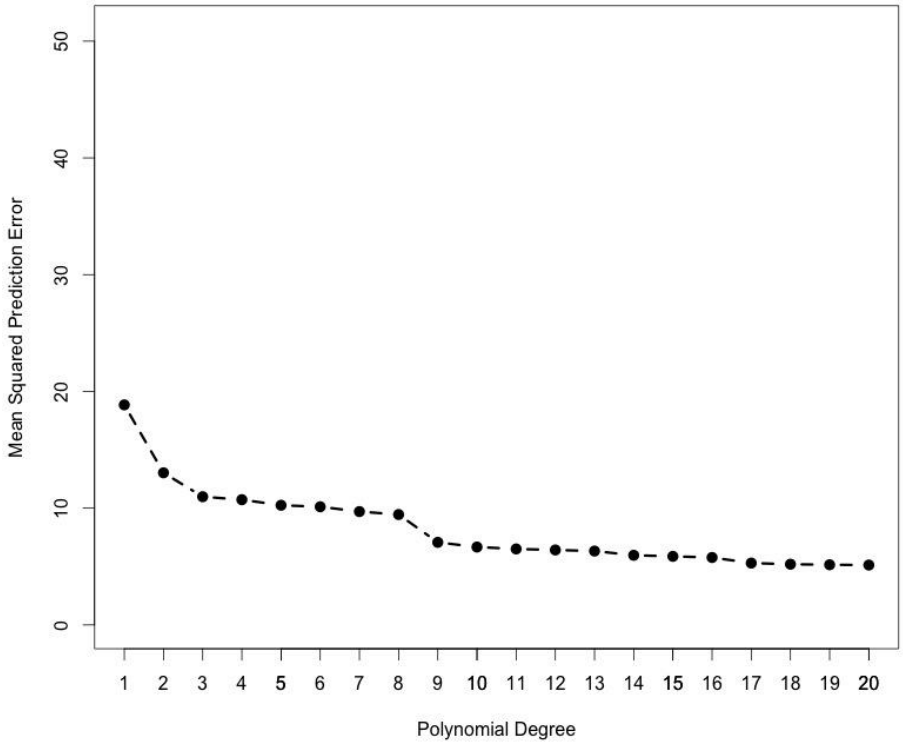


Polynomial Curve-Fitting Example

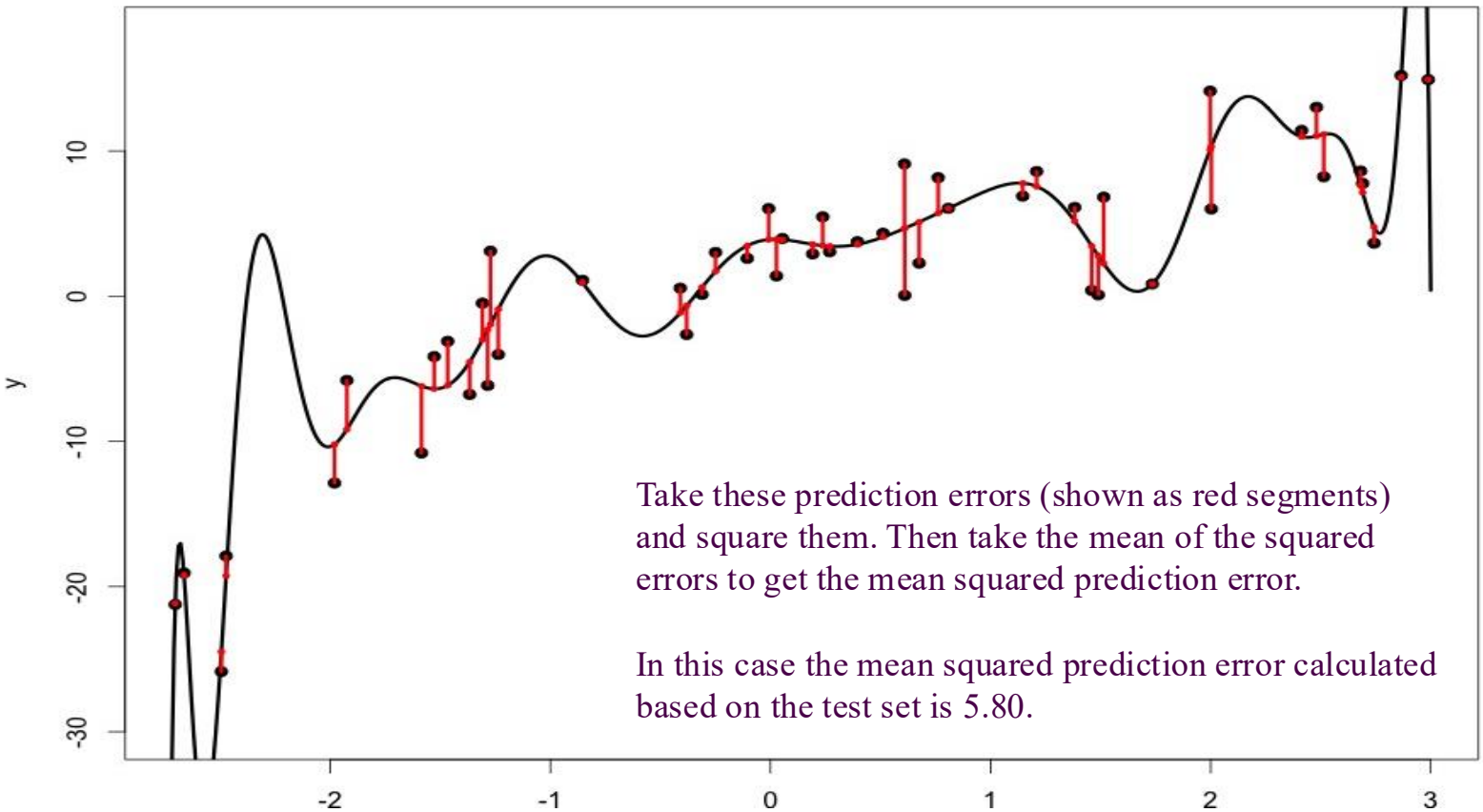


Training Set Prediction Error

The prediction error calculated on the training set decreases in lock-step as model complexity increases. The 20th degree polynomial model fits the best according to the training data.



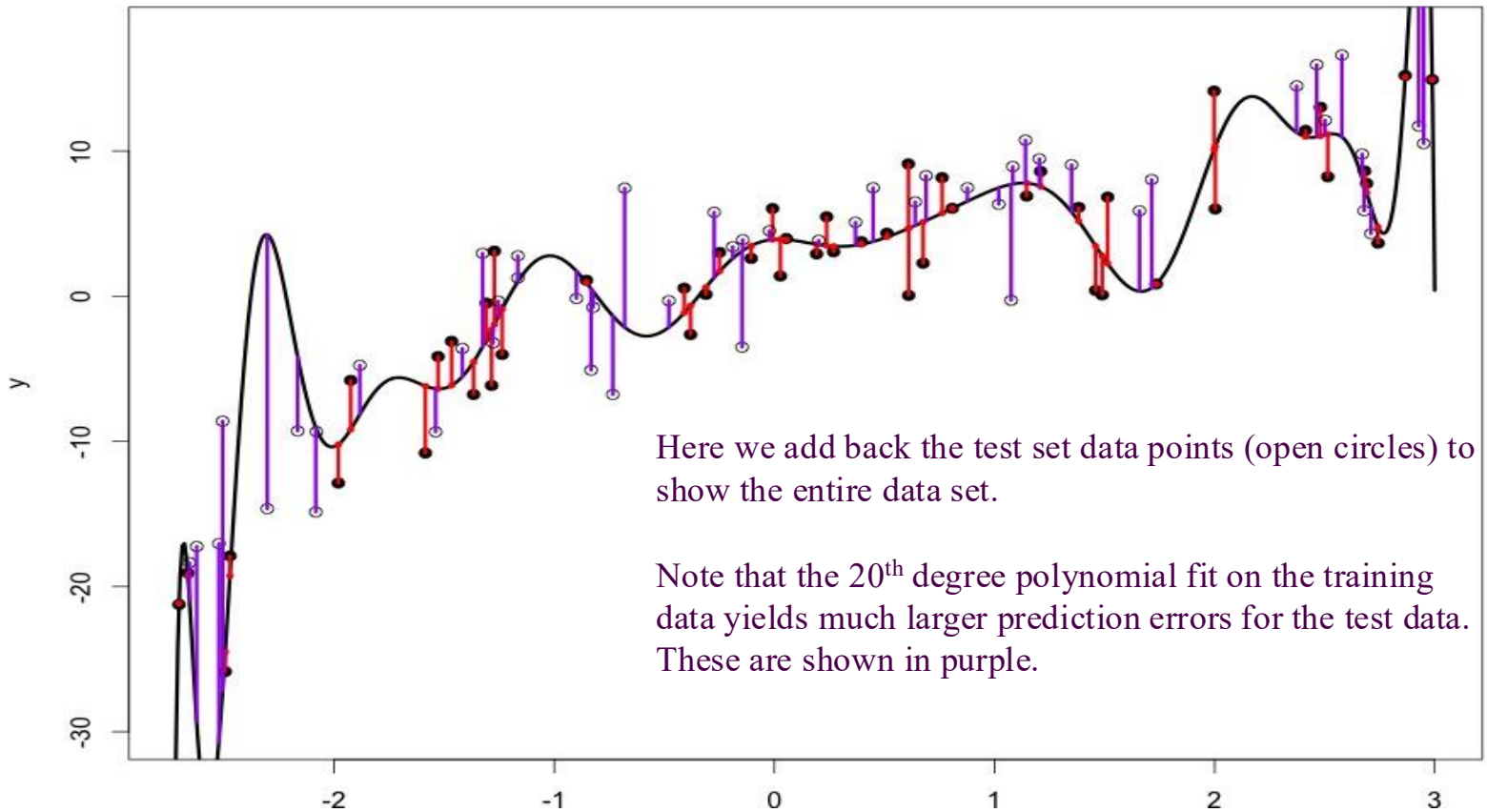
Training Set Prediction Error (Degree = 20)



Take these prediction errors (shown as red segments) and square them. Then take the mean of the squared errors to get the mean squared prediction error.

In this case the mean squared prediction error calculated based on the test set is 5.80.

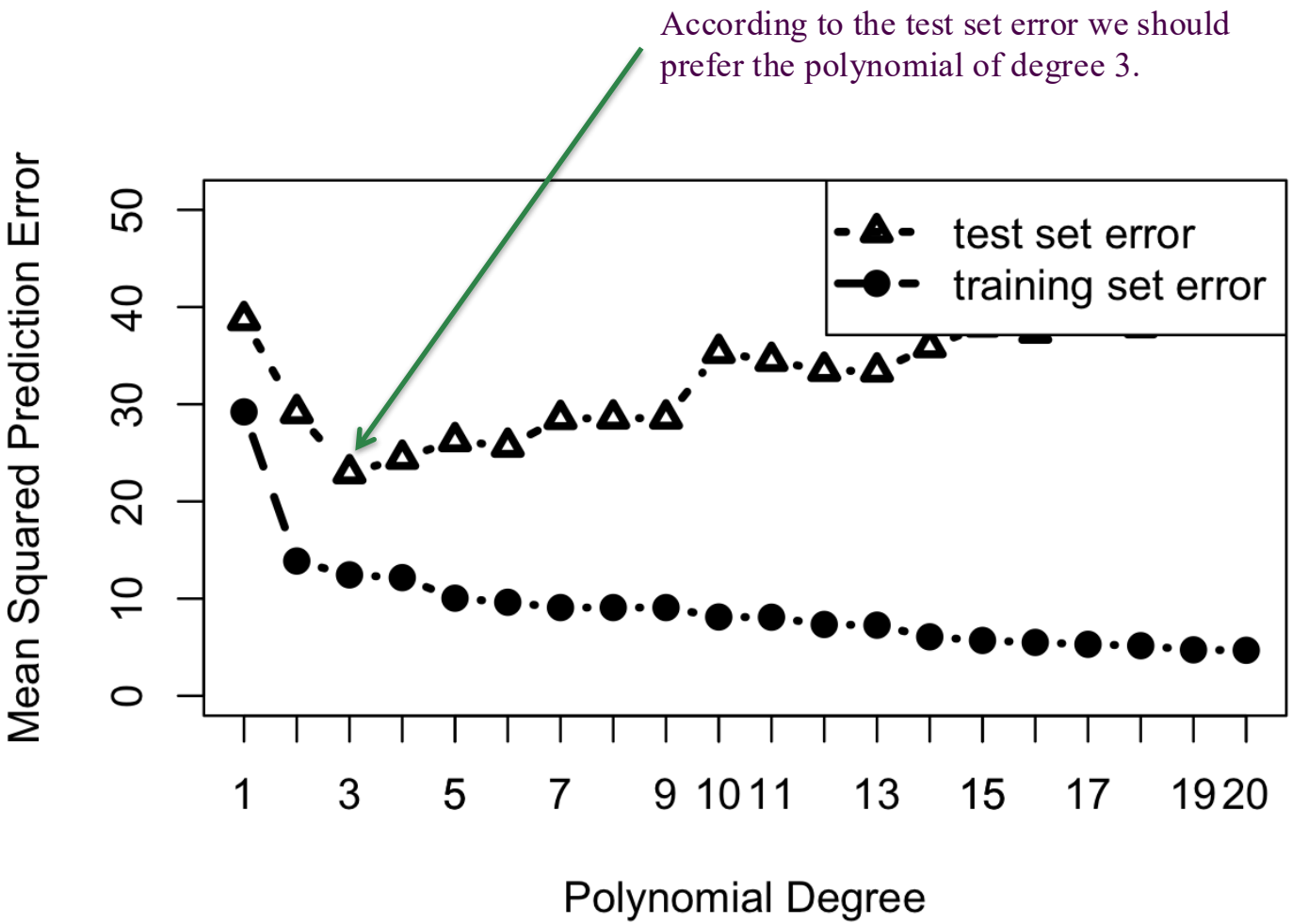
Training Set Prediction Error (Degree = 20)



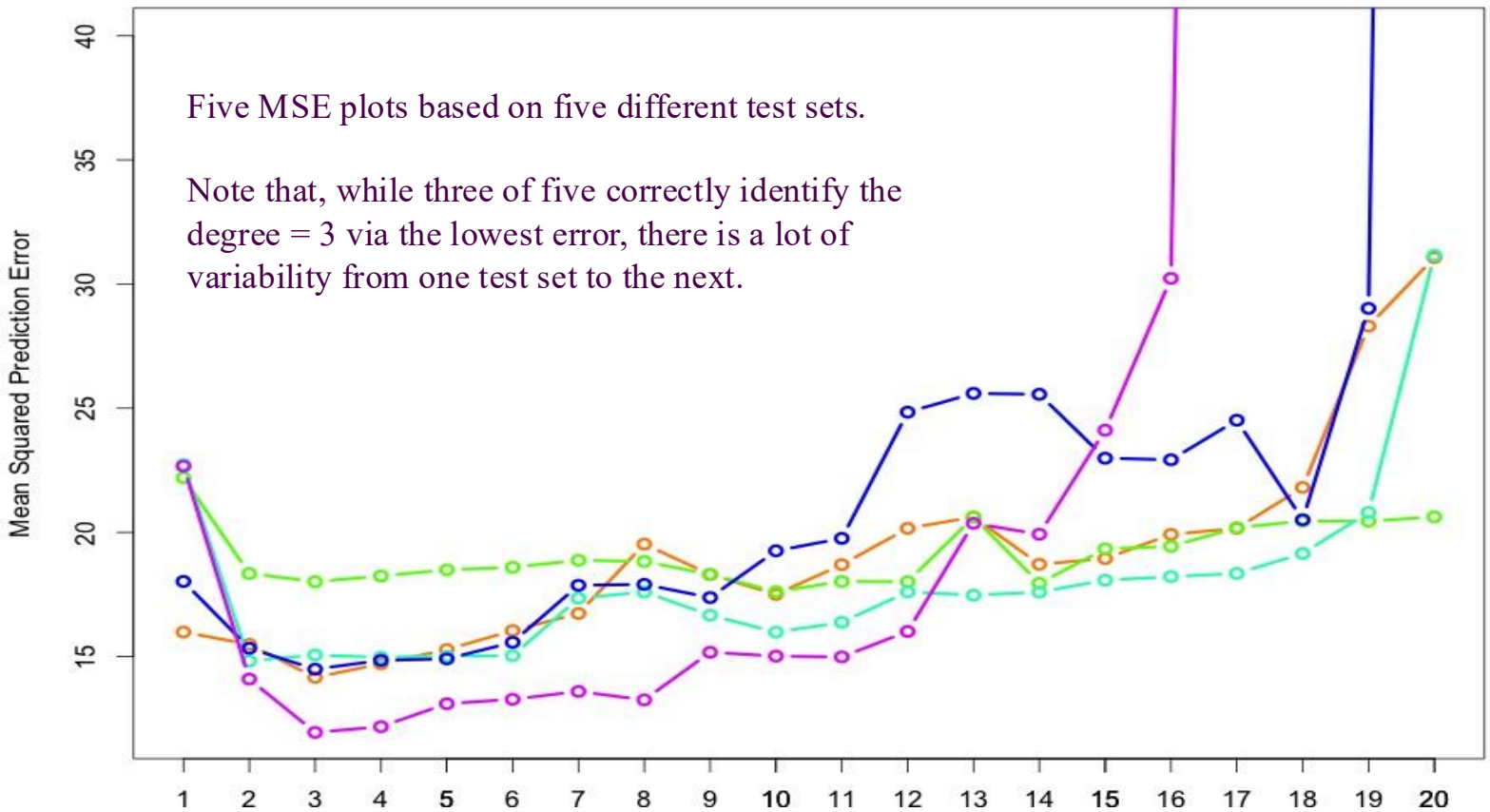
Here we add back the test set data points (open circles) to show the entire data set.

Note that the 20th degree polynomial fit on the training data yields much larger prediction errors for the test data. These are shown in purple.

Training and Test Set Prediction Error



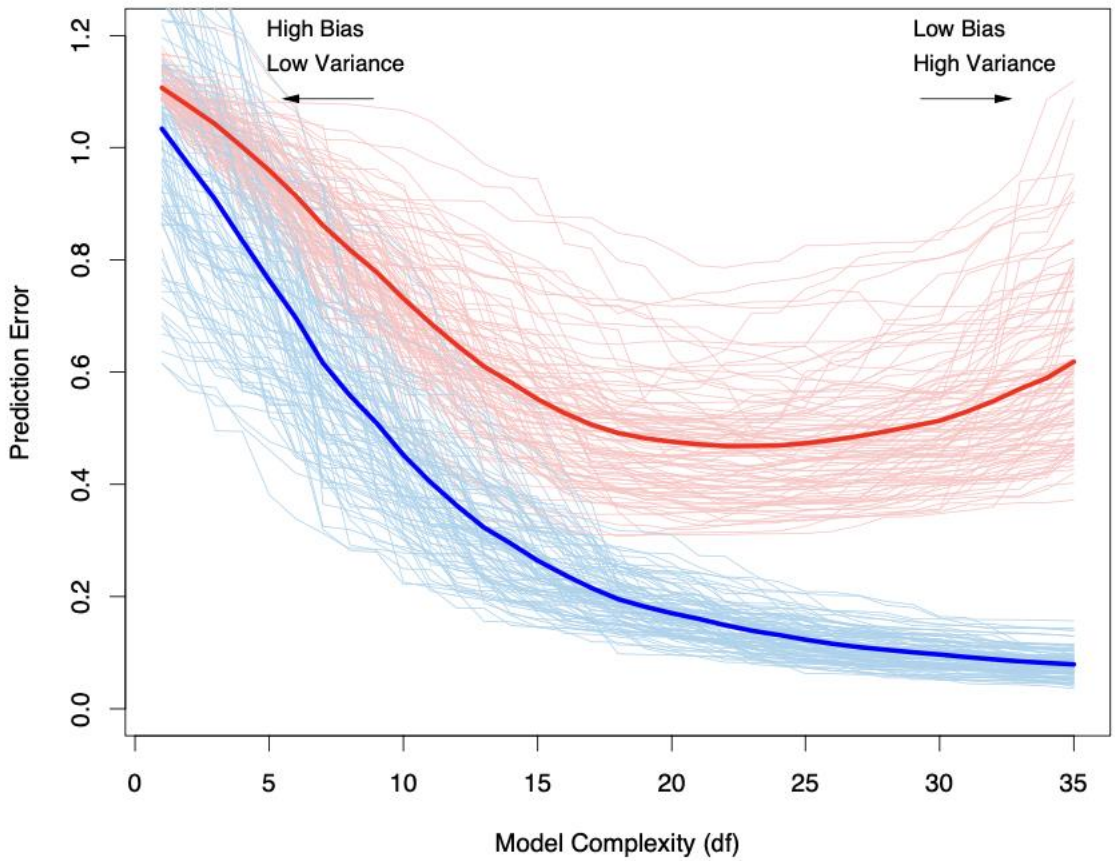
Variability Based on Random Selection of Test Set



Five MSE plots based on five different test sets.

Note that, while three of five correctly identify the degree = 3 via the lowest error, there is a lot of variability from one test set to the next.

Back to the general graph



Behavior of test sample and training sample error as the model complexity is varied.
Source: “The Elements of Statistical Learning” by Hastie, Tibshirani and Friedman

Bias – variance tradeoff (again)

- In regression settings there is a similar result that explains the tradeoff between bias and variance.
- What is the ”true value of the parameter” aka *ideal predictor*?

$$\eta(x) = E(Y|X = x)$$

- What is the estimator aka *learned predictor*?

$$\hat{f}(x) = \underset{f \in \mathcal{F}}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - f(x_i)\right)^2$$

- What is the MSE aka *average learning error*?

$$E\left[\left(\hat{f}(x) - \eta(x)\right)^2\right] = \underbrace{\left(E\left[\hat{f}(x)\right] - \eta(x)\right)^2}_{Bias^2} + \underbrace{E\left[\left(\hat{f}(x) - E\left[\hat{f}(x)\right]\right)^2\right]}_{Variance}$$

Bias – variance tradeoff (again)

- **Bias squared:**

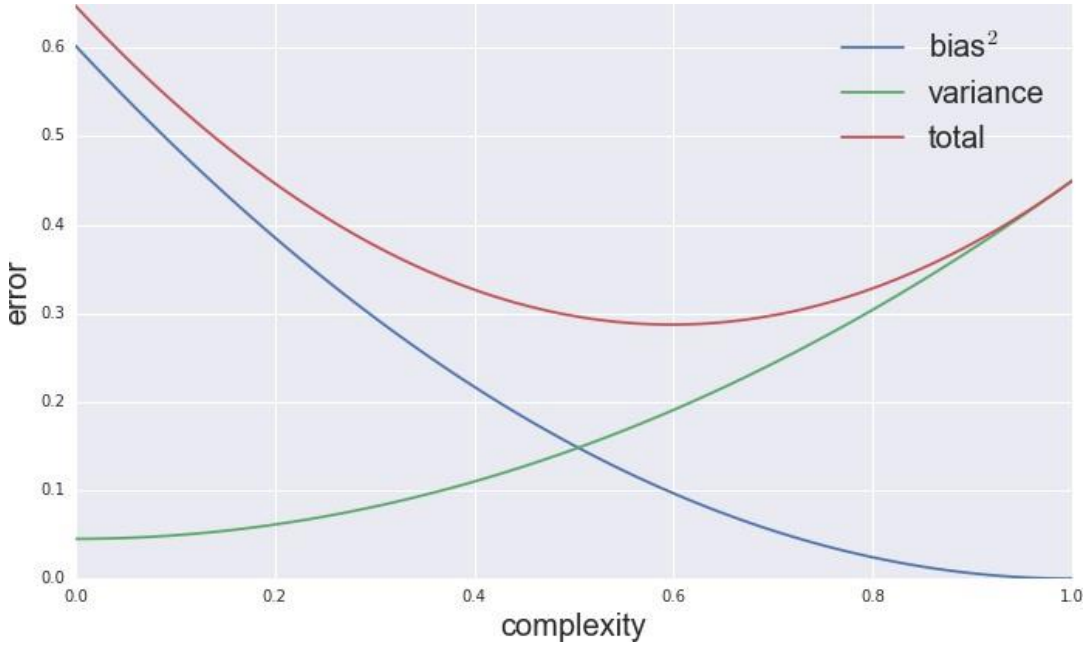
measures how the predictor is mismatched with the best predictor in expectation

- **Variance:**

measures how the predictor varies each time with a new training datasets

- **Test error rate:**

Essentially estimates the MSE (or total) without using formulas.

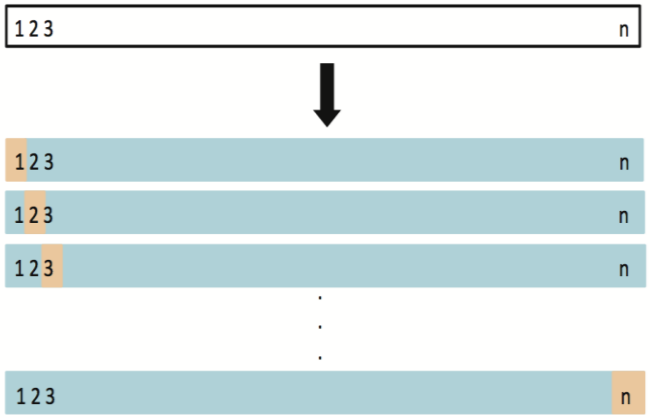


Problems With a Single Validation Set

- Beyond the variability identified in the last plot, another concern is that only a subset of the observations are used in the model training.
- In our case, we used 50 instead of 100 observations to fit the model.
- Most statistical methods perform more poorly when they are trained on fewer observations, so use of a single validation set may lead to overestimation of the error rate.
- *Cross-validation* is an extension of the validation set approach we just discussed that can address both issues:
  1. Cross-validation averages error rates over multiple samples (i.e., *resampling*) to address the variability of MSEs.
  2. Cross-validation fits may be carried out with as many as all but one of the observations in a data set, thereby minimizing the extent to which we overestimate error rate due to lack of sufficient sample size relative to the full sample.

Leave-One-Out Cross-Validation

- Instead of creating two subsets of comparable size, a single observation is used for the validation set and the remaining observations (n – 1) make up the training set.

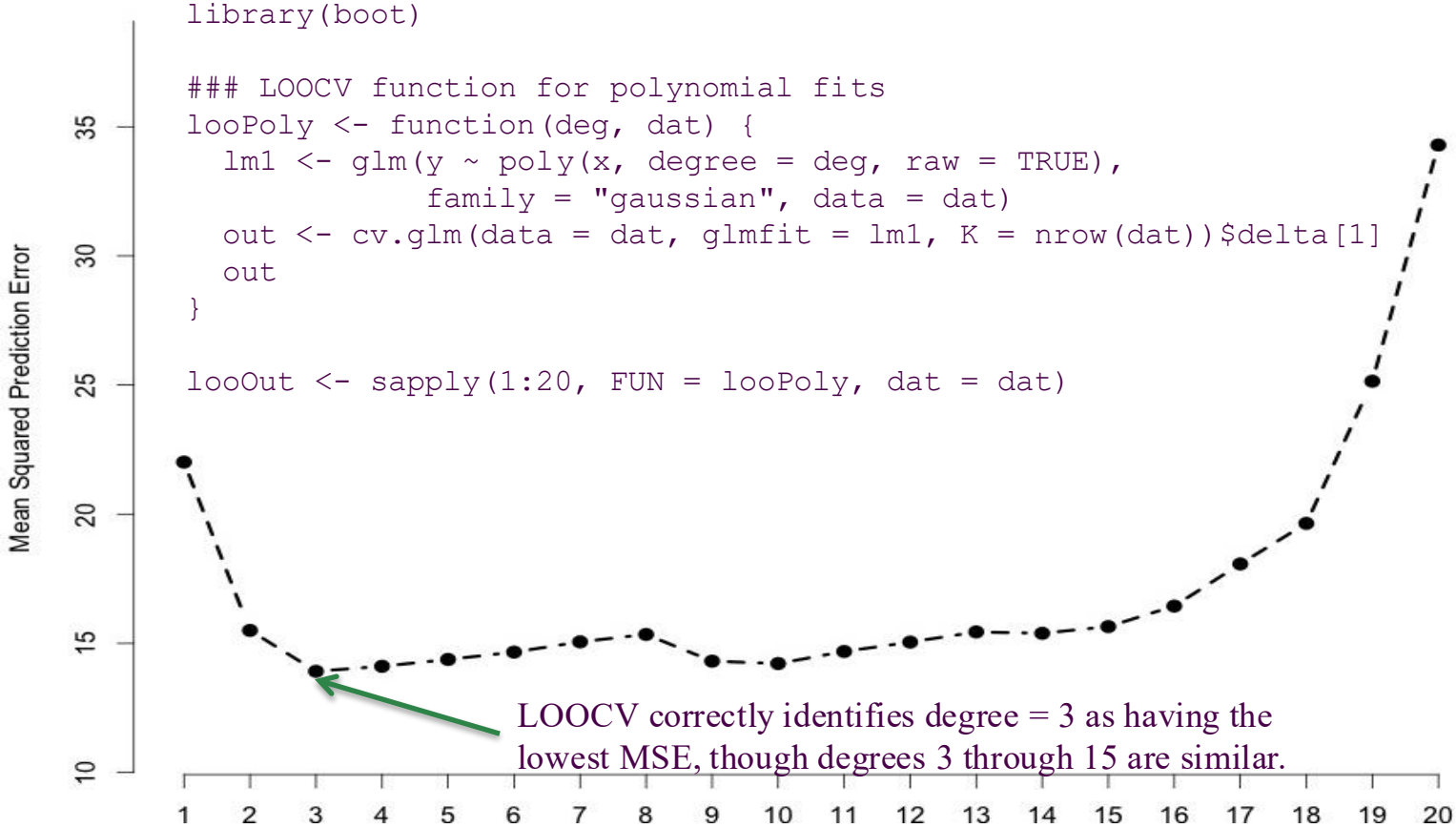


$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

- **LOOCV Algorithm:**
  - Split the entire data set of size n into:
    - Blue = training data set
    - Beige = validation data set
  - Fit the model using the training data set
  - Evaluate the model using validation set and compute the corresponding MSE.
  - Repeat this process n times, producing n squared errors. The average of these n squared errors estimates the test MSE.

LOOCV MSEs

```
### LOOCV with package boot
library(boot)

### LOOCV function for polynomial fits
looPoly <- function(deg, dat) {
  lm1 <- glm(y ~ poly(x, degree = deg, raw = TRUE),
             family = "gaussian", data = dat)
  out <- cv.glm(data = dat, glmfit = lm1, K = nrow(dat))$delta[1]
  out
}

looOut <- sapply(1:20, FUN = looPoly, dat = dat)
```



Validation Set Approach vs. LOOCV

- LOOCV has far less bias and, therefore, tends not to overestimate the test error rate.

- Performing LOOCV multiple times always yields the same results because there is no randomness in the training/validation set splits.

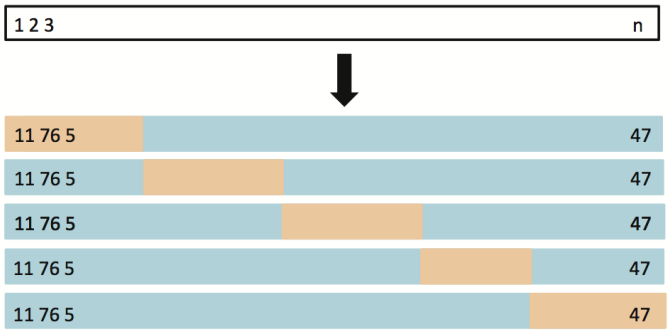
- LOOCV is computationally intensive because the model has to be fit n times. However, there is a shortcut with OLS linear or polynomial regression (where h_i is the leverage):

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{Y_i - \hat{Y}_i}{1 - h_i}\right)^2$$

K-Fold Cross-Validation

- Probably the simplest and most widely used method for estimating prediction error.
- This method directly estimates the average prediction error when the machine learning method is applied to an independent test sample.
- Ideally, if we had enough data, we would set aside a validation set (as previously described) and use it to assess the performance of our prediction model.
- To finesse the problem, *K*-fold cross-validation uses part of the available data to fit the model, and a different part to test it.

K-Fold Cross-Validation (cont.)



- We use this method because LOOCV is computationally intensive/slow.
- We randomly divide the data set of into K folds (typically $K = 5$ or 10).
- One of the folds is treated as a test set, and the method is fit on the remaining $K - 1$ folds. The MSPE is computed on the observations in the *held-out* fold. The process is repeated K times, taking out a different part each time.
- By averaging the K estimates of the test error, we get an estimated validation (test) error rate for new observations.

K-Fold Cross-Validation (cont.)

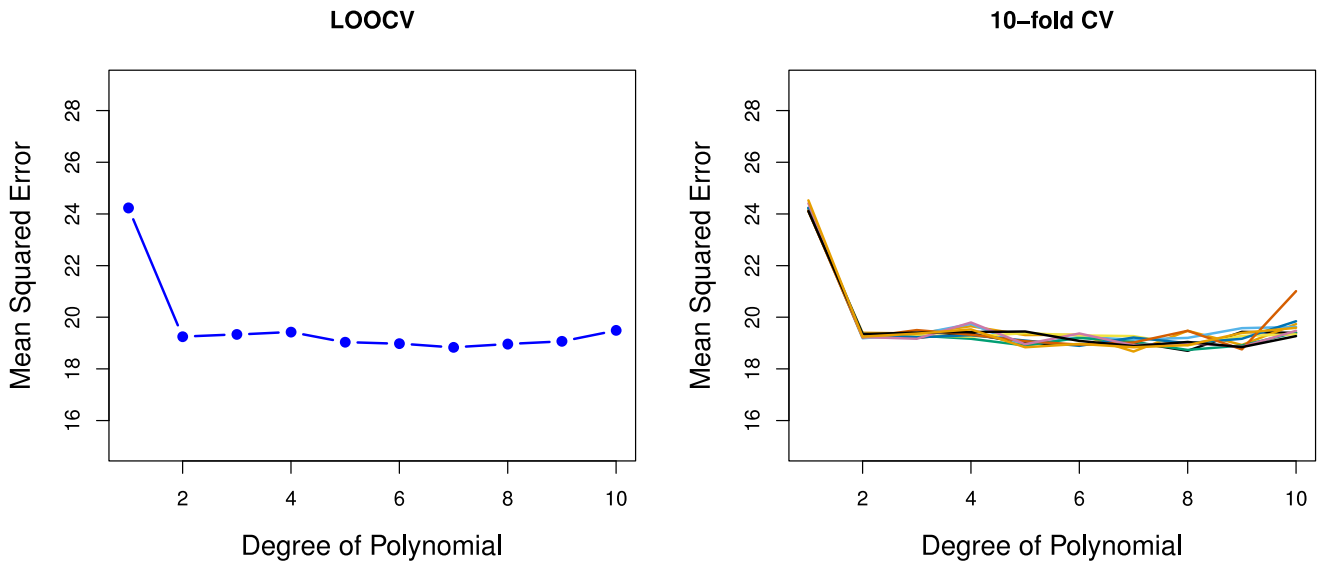
- Let the K folds be $C_1, \ldots, C_K$, where C_k denotes the indices of the observations in fold k. There are n_k observations in fold k: if N is a multiple of K, then $n_k = n / K$.

- Compute: $\mathrm{CV}_{(K)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{MSE}_k$

where $\mathrm{MSE}_k = \frac{1}{n_k} \sum_{i \in C_k} (Y_i - \hat{Y}_i)^2$ and $\hat{Y}_i$ is the fitted value for observation i, obtained from the data with fold k removed.

1	2	3	4	5
Train	Train	Validation	Train	Train

K-Fold Cross-Validation vs. LOOCV



- **Left Panel:** LOOCV Error Curve
- **Right Panel:** *10*-fold CV run nine separate times, each with a different random split of the data into ten parts.
- **Note:** LOOCV is a special case of K-fold, where $K = n$

Bias-Variance Trade-off for *K*-Fold Cross-Validation

- Which is better, LOOCV or *K*-fold CV?
  - LOOCV is more computationally intensive than *K*-fold CV
  - From the perspective of bias reduction, LOOCV is preferred to *K*-fold CV (when $K < n$)
  - However, LOOCV has higher variance than *K*-fold CV (when $K < n$)
  - Thus, we see the bias-variance trade-off between the two resampling methods

- We tend to use *K*-fold CV with $K = 5$ or $K = 10$, as these values have been shown empirically to yield test error rate estimates that suffer neither from excessively high bias nor from very high variance

Cross-Validation on Classification Problems

- We will cover classification problems in more detail later in the course, but we briefly show how CV can be used when Y is qualitative (categorical) as opposed to quantitative. Here, rather than use MSE to quantify test error, we instead use the number of misclassified observation.

- LOOCV Error Rate: $\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{Err}_i$, where $\mathrm{Err}_i = I(Y_i \neq \hat{Y}_i)$

- We use CV as follows:
  - Divide data into K folds; hold-out one part and fit using the remaining data (compute error rate on hold-out data); repeat K times.
  - CV Error Rate: average over the K errors we have computed

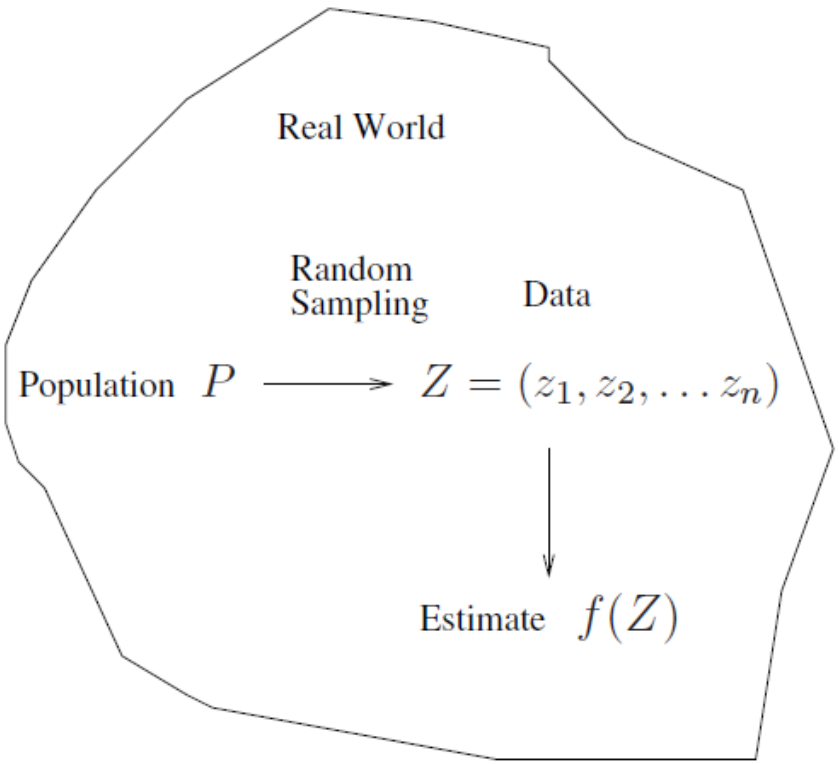
The Bootstrap

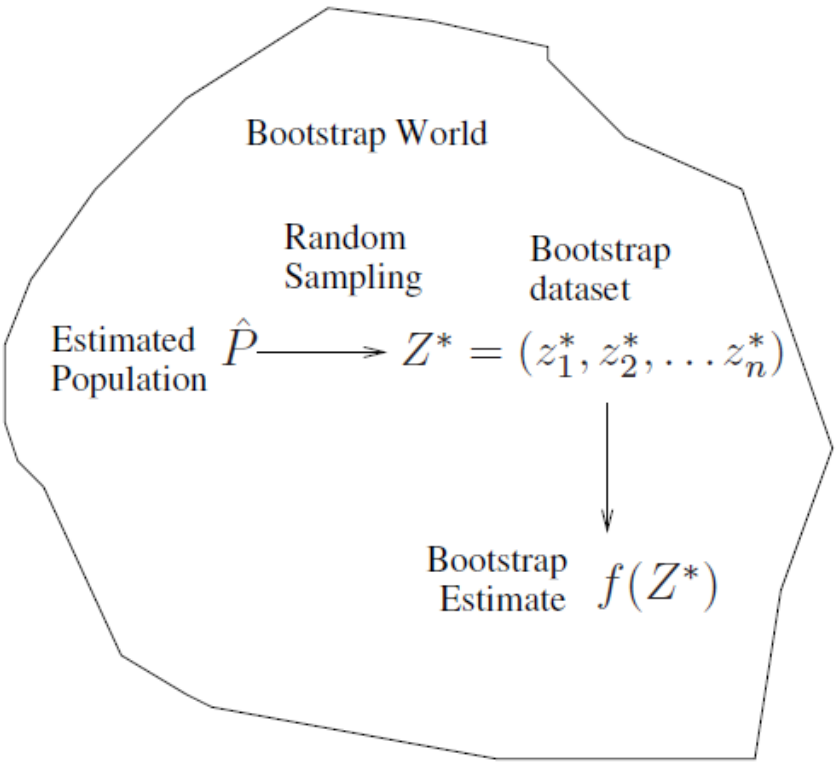
- The *bootstrap* is a flexible and powerful statistical tool that can be used to quantify uncertainty associated with a given estimator or statistical method; it is a general tool for assessing statistical accuracy.

- As an example, the bootstrap can be used to estimate the standard errors of the coefficients from a linear regression fit, or a confidence interval for that coefficient.

- The use of the term bootstrap derives from the phrase "to pull oneself up by one's bootstraps."

The Bootstrap: Overview





The Bootstrap: Overview (cont.)

- Suppose we have a model fit to a set of training data. We denote the training set by $\mathbf{Z} = (z_1, z_2, \ldots, z_N)$ where $z_i = (x_i, y_i)$.

- The basic idea is to randomly draw datasets **with replacement** from the training data, each sample the same size as the original training set.

- This is done B times, producing B bootstrap datasets. Then we refit the model to each of the bootstrap datasets, and examine the behavior of the fits over the B replications.

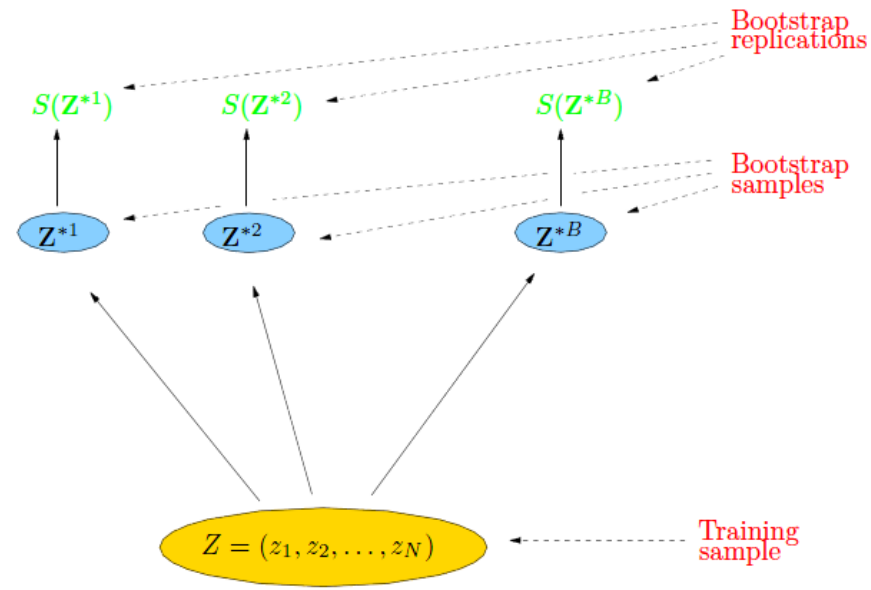
The Bootstrap: Overview (cont.)

- $S(\mathbf{Z})$ is any quantity computed from the data $\mathbf{Z}$, for example, the prediction at some input point.

- From the bootstrap sampling we can estimate any aspect of the distribution of $S(\mathbf{Z})$, for example, its variance:

$$\widehat{\mathrm{Var}}[S(\mathbf{Z})] = \frac{1}{B-1}\sum_{b=1}^{B}(S(\mathbf{Z}^{*b}) - \bar{S}^*)^2$$

$$\bar{S}^* = \sum_b S(\mathbf{Z}^{*b})/B$$



Note that this estimated variance can be thought of as a Monte-Carlo estimate of the variance of $S(\mathbf{Z})$ under sampling from the empirical distribution function.

The Bootstrap: An Example

- Suppose that we wish to invest a fixed sum of money in two financial assets that yield returns X and Y; note that X and Y are random variables.

- We will invest a fraction α of our money in X, and will invest the remaining $1 - \alpha$ in Y.

- **Goal:** Since there is variability associated with the returns on these two assets, we wish to choose α to minimize the total risk, or variance, of our investment.

The Bootstrap: An Example (cont.)

- In other words, we want to minimize $\mathrm{Var}(\alpha X + (1-\alpha)Y)$

- The value that minimizes the risk, in this examples, is: $\alpha = \dfrac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$

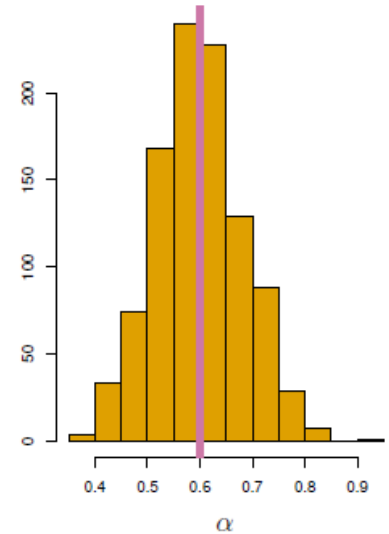
where $\sigma_X^2 = \mathrm{Var}(X), \sigma_Y^2 = \mathrm{Var}(Y)$, and $\sigma_{XY} = \mathrm{Cov}(X, Y)$

- In reality, the quantities σ_X^2, σ_Y^2, and σ_{XY} are unknown, so we can compute estimates for these quantities using a data set that contains past measurements for X and Y. We can then estimate the value of α that minimizes risk using:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}$$

The Bootstrap: An Example (cont.)

- We can use this approach for estimating α on a simulated data set, where 100 pairs of returns for the investments X and Y are simulated.



- In order to quantify the accuracy of our estimate of α, we also estimate the standard deviation of $\hat{\alpha}$.

- To do so, the process of simulating 100 paired observations of X and Y is repeated 1,000 times; now we have 1,000 estimates for α.

- The mean over all 1,000 estimates for α → $$\bar{\alpha} = \frac{1}{1000}\sum_{r=1}^{1000}\hat{\alpha}_r = 0.5996$$

- The standard deviation of the estimates → $$\sqrt{\frac{1}{1000-1}\sum_{r=1}^{1000}(\hat{\alpha}_r - \bar{\alpha})^2} = 0.083$$

The Bootstrap: An Example (cont.)

- In rough terms, for a random sample from the population, we would expect $\hat{\alpha}$ to differ from α by approximately 0.08, on average.

- In the real world, this procedure **cannot be applied** because for real data we cannot generate new samples from the original population.

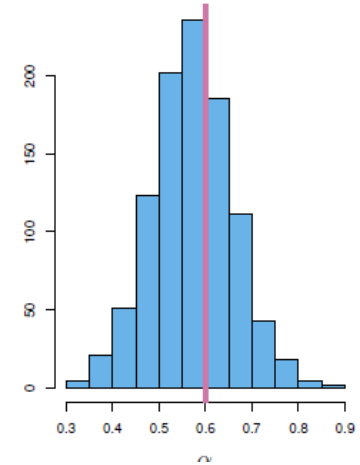
- The **bootstrap** approach, however, allows us to use a computer to mimic the process of obtaining new data sets; this enables use to estimate the variability of our estimate without generating additional samples.

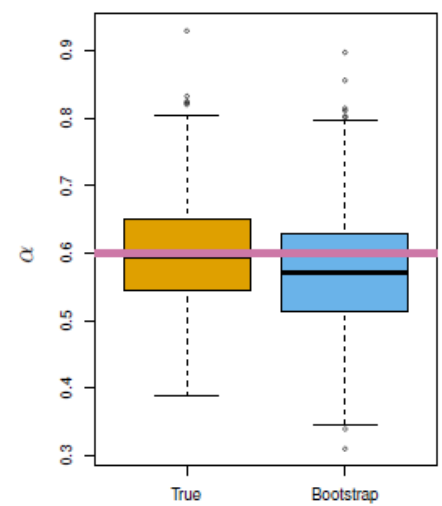
The Bootstrap: An Example (cont.)

- Rather than repeatedly obtaining independent data sets from the population, we instead obtain distinct data sets by repeatedly sampling observations from the original data set **with replacement**.

- Each of these *bootstrap data sets* is created by sampling *with replacement*, and is the *same size* as the original data set.

- As a result, some observations may appear more than once in a given bootstrap data set and some not at all.



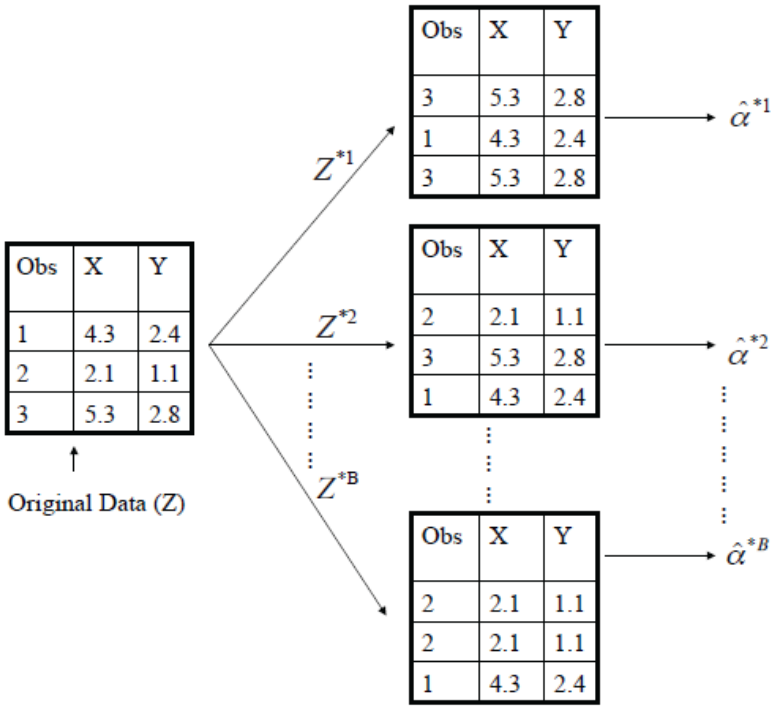


The Bootstrap: An Example (cont.)

- A graphical illustration of the bootstrap approach on a small sample containing $n = 3$ observations.

- Each bootstrap data set contains n observations, sampled **with replacement** from the original data set.

- Each bootstrap data set is used to obtain an estimate of α.



The Bootstrap: An Example (cont.)

- Denoting the first bootstrap data set by Z^{*1}, we can use Z^{*1} to produce a new bootstrap estimate for α, which we call $\hat{\alpha}^{*1}$.
- This procedure is repeated B times for some large value of B (say 100 or 1000), in order to produce B different bootstrap data sets, Z^{*1}, Z^{*2},…, Z^{*B}, and B corresponding α estimates, $\hat{\alpha}^{*1}$, $\hat{\alpha}^{*2}$,…, $\hat{\alpha}^{*B}$.
- We compute the standard error of these bootstrap estimates using:

$$\mathrm{SE}_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \bar{\hat{\alpha}}^{*}\right)^2}$$

- This serves as an estimate of the standard error of $\hat{\alpha}$ estimated from the original data set.

The Bootstrap: More Details

- Although the bootstrap is used primarily to obtain standard errors of an estimate, it can also provide approximate confidence intervals for a population parameters.
  - One approach is known as *Bootstrap Percentile* confidence interval.

- In cross-validation, each of the K validation folds is distinct from the other $K - 1$ folds used for training (i.e. there is no overlap).

- To estimate the prediction error using the bootstrap, one approach would be to fit the model in question on a set of bootstrap samples, and then keep track of how well it predicts the original training set.

The Bootstrap: More Details (cont.)

- If $\hat{f}^{*b}(x_i)$ is the predicted value at x_i, from the model fitted to the bth bootstrap dataset, our estimate is:

$$\widehat{\mathrm{Err}}_{\mathrm{boot}} = \frac{1}{B}\frac{1}{N}\sum_{b=1}^{B}\sum_{i=1}^{N} L(y_i, \hat{f}^{*b}(x_i))$$

- This estimate does not provide a good estimate in general because the bootstrap datasets are acting as the training samples, while the original training set is acting as the test sample, and these two samples have observations in common.
  - Each bootstrap sample has significant overlap with the original data.

The Bootstrap: More Details (cont.)

- Note that about 2/3 of the original N data points appear in each bootstrap sample:
  - $\Pr\{\text{observation } i \in \text{bootstrap sample } b\} = 1 - (1 - \frac{1}{N})^N \approx 1 - e^{-1} = 0.632$
- This overlap can make overfit predictions like unrealistically good, and is the reason that cross-validation explicitly uses non-overlapping data for the training and test samples.
- In other words, this will cause the bootstrap to seriously *underestimate* the true prediction error.
- By mimicking cross-validation, a better bootstrap estimate can be obtained.

The Bootstrap: Exercise

- Suppose we want to use the sample median as an estimator of population median.
- What is its standard error?
- No exact formula exists!
- There is an asymptotic result:
- Suppose that we have a sample of size $n = 2m + 1$ from a population with a continuous distribution whose pdf is $f(x)$. Furthermore, suppose that $f(x) \neq 0$ for $x =$ the population median. For n sufficiently large, the sampling distribution of the median (denoted μ-tilde) is approximately normal with
- $mean = \tilde{\mu}$
- $variance = \frac{1}{8mf^2(\tilde{\mu})}$
- This means that the asymptotic standard error is

$$\mathrm{SE}(\tilde{\mu}) = \frac{1}{2mf(\tilde{\mu})\sqrt{2m}}$$

- Use a simulation from Exponential distribution and apply bootstrap to obtain $\mathrm{SE}(\tilde{\mu})$